

ER to Relational Mapping

COSC 304 – Introduction to Database Systems



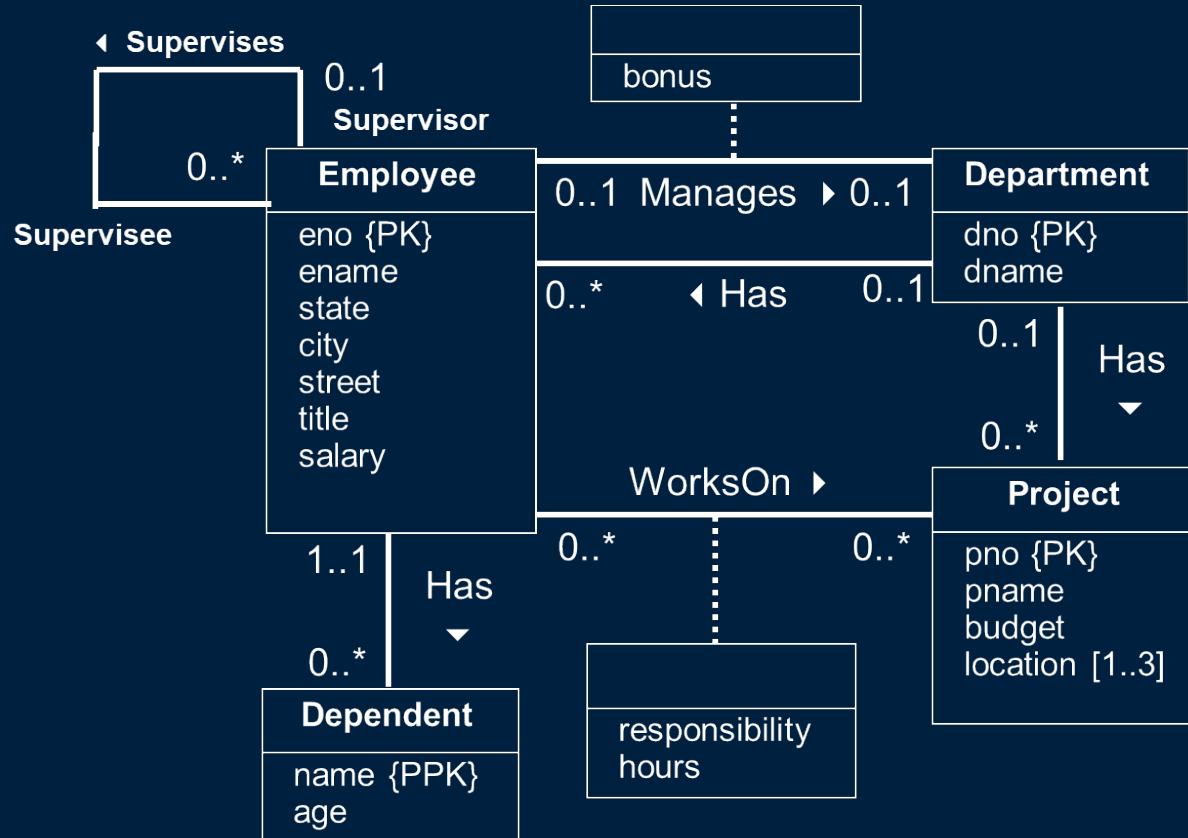
ER Model to Relational Schemas

Converting an ER model to a relational database schema involves 7 steps. An additional eighth step handles EER superclass/subclass structures.

In general:

- entity types become relations
- M:N and n-ary relationships become relations
- 1:1 and 1:N relationships are usually represented using foreign keys

ER to Relational Example

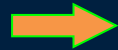
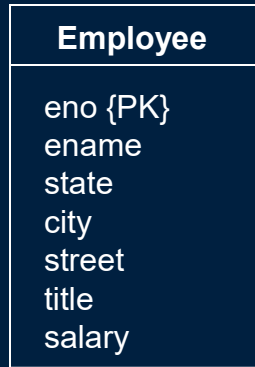


ER to Relational Mapping

Step #1: Convert Strong Entities



Step #1: Convert each strong entity type to a relation.



Employee (eno, ename, state, city, street, title, salary)

- Notes:

- Entity attributes become relation attributes.
- Entity key becomes the relation's primary key.
- Multivalued attributes are handled separately in Step 6.
- Composite attributes are represented by their component attributes.
- Derived attributes are normally not stored.

ER to Relational Mapping

Current Relational Schema - Step #1



Employee (eno, ename, state, city, street, title, salary)

Project (pno, pname, budget)

Department (dno, dname)

ER to Relational Mapping

Step #2: Convert Weak Entities



Step #2: Convert each weak entity into a relation with foreign keys to its identifying relations (entities). Example:



Employee (eno, ename, state, city, street, title, salary)

Include the weak entity's attributes.

Add the primary key(s) of its owner entity/entities as a foreign keys.

The primary key is the owner key(s) + any partial key attributes.



Dependent (eno, name, age)

Dependent.eno is a FK to Employee.eno

ER to Relational Mapping

Current Relational Schema - Step #2



Dependent (eno, name, age)



Employee (eno, ename, state, city, street, title, salary)

Project (pno, pname, budget)

Department (dno, dname)

ER to Relational Mapping

Steps #3-5: Convert Relationships



Steps #3 to #5 convert *binary* relationships of cardinality:

- 1:1 - Step #3 - UNIQUE FOREIGN KEY
- 1:N - Step #4 - FOREIGN KEY on N-side
- M:N - Step #5 – create new relation

The M:N approach of creating a separate relationship relation can also represent 1:1 and 1:N relationships. However, for 1:1 and 1:N relationships, a foreign key usually gives a simpler relational schema without an unnecessary relation.

Non-binary relationships are handled in Step #7.

ER to Relational Mapping

Step #3: Convert 1:1 Relationships

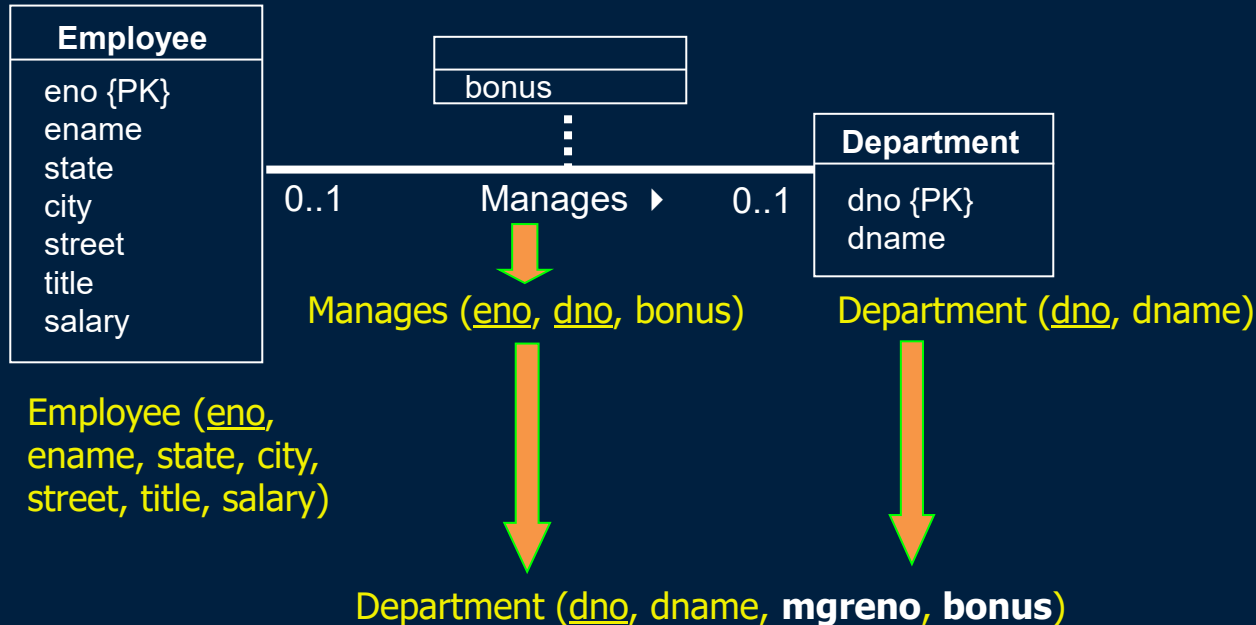


Step #3: Convert binary 1:1 relationships by placing a `UNIQUE` foreign key in one of the two relations.

- Choose one of the two relations.
- Add the other relation's primary key as a `UNIQUE` foreign key.
- Add any relationship attributes there as well.
- Prefer the side with mandatory participation, or the side expected to participate most often.

ER to Relational Mapping

Step #3: Convert 1:1 Relationships (2)



Note: Renamed eno to mgreno for clarity.
mgreno has a UNIQUE constraint and is a foreign key to Employee.

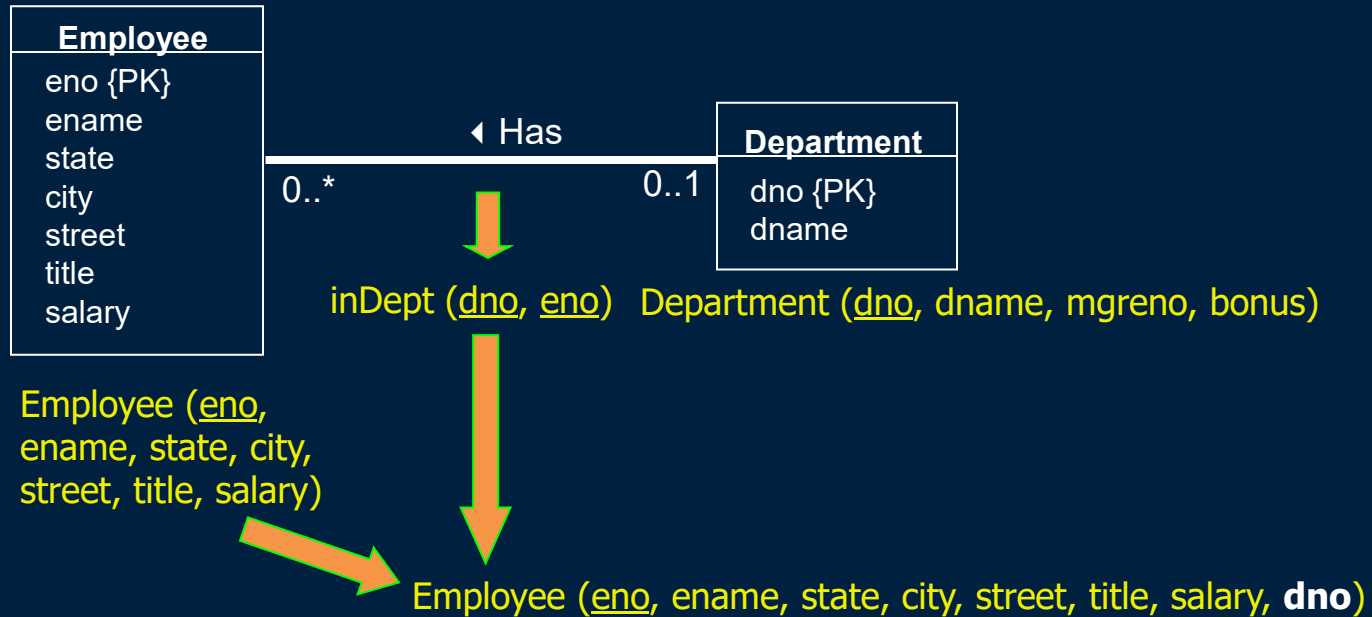
Step #4: Convert 1:N Relationships

Step #4: Convert binary 1:N relationships by adding the primary key of the 1-side as a foreign key in the N-side relation. Any relationship attributes are also added to the N-side relation.

Unlike 1:1 relationships, there is no choice: the foreign key goes on the N side.

ER to Relational Mapping

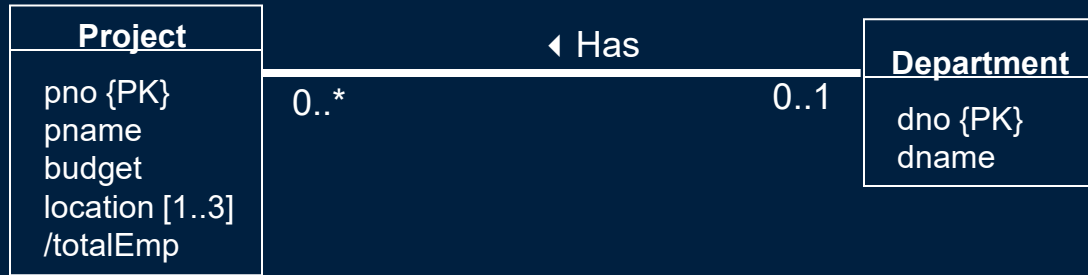
Step #4: Convert 1:N Relationships



Employee.dno is a foreign key to Department.

ER to Relational Mapping

Step #4: Convert 1:N Relationships



Question: What table has the foreign key field?

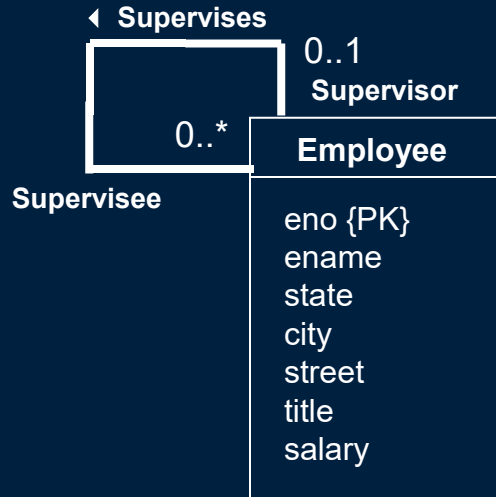
A) Project

B) Department

C) None

ER to Relational Mapping

Step #4: Convert 1:N Relationships



→ Supervises (supereno, eno)

Employee (eno, ename, state, city, street, title, salary, dno)

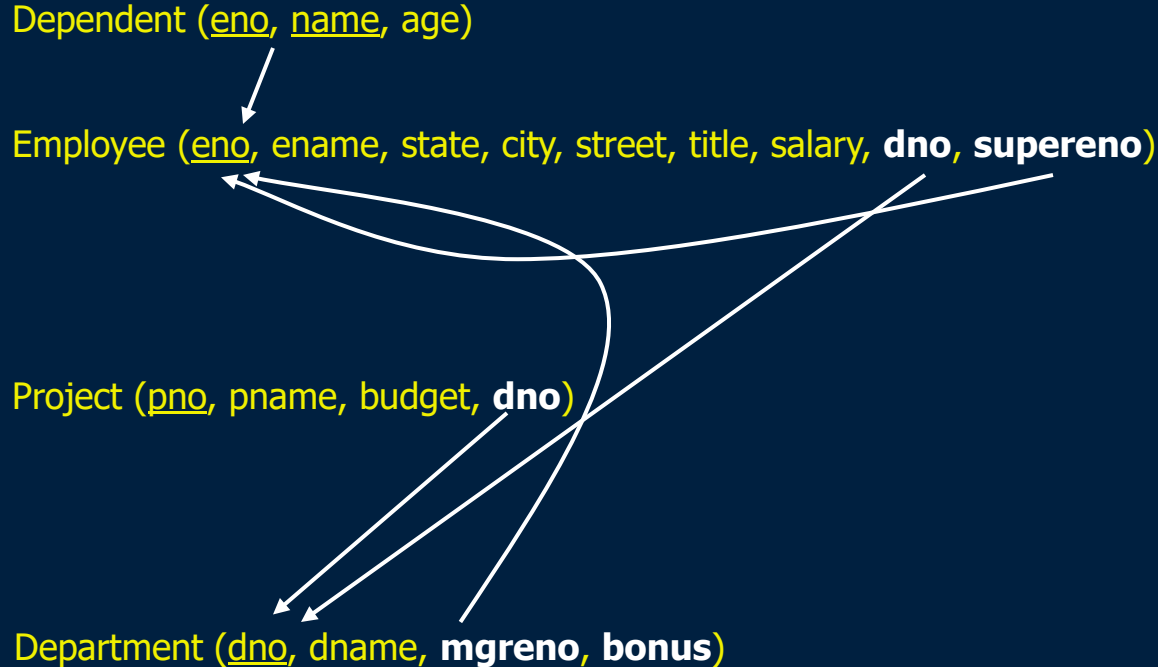


Employee (eno, ename, state, city, street, title, salary, dno, **supereno**)

Employee.supereno is a self-referencing foreign key to Employee.eno.

ER to Relational Mapping

Current Relational Schema - Step #4

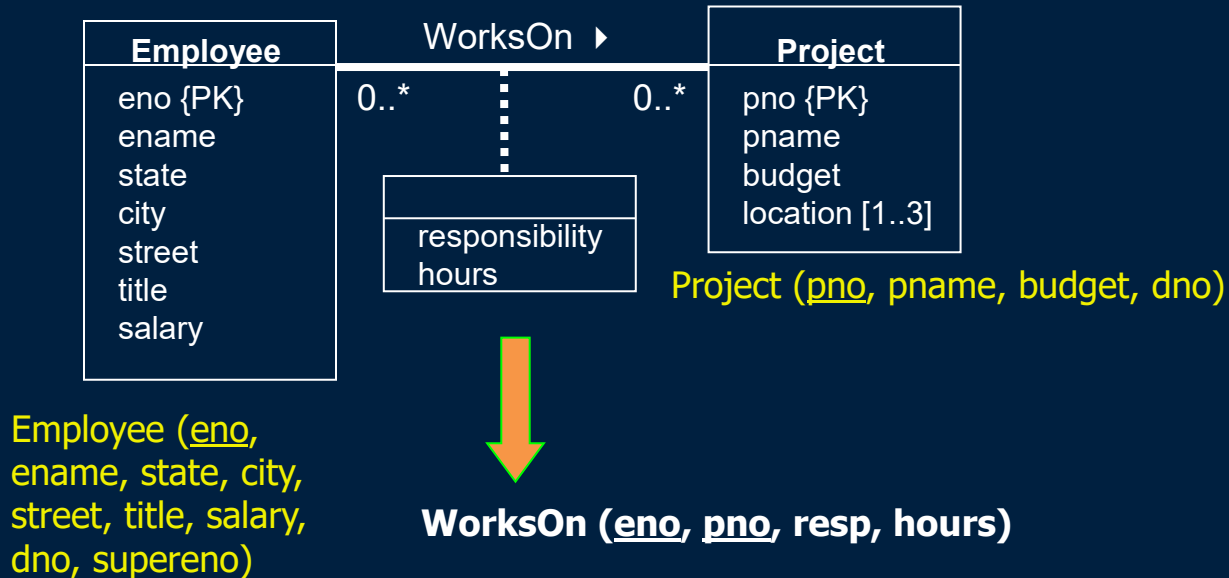


ER to Relational Mapping

Step #5: Convert M:N Relationships

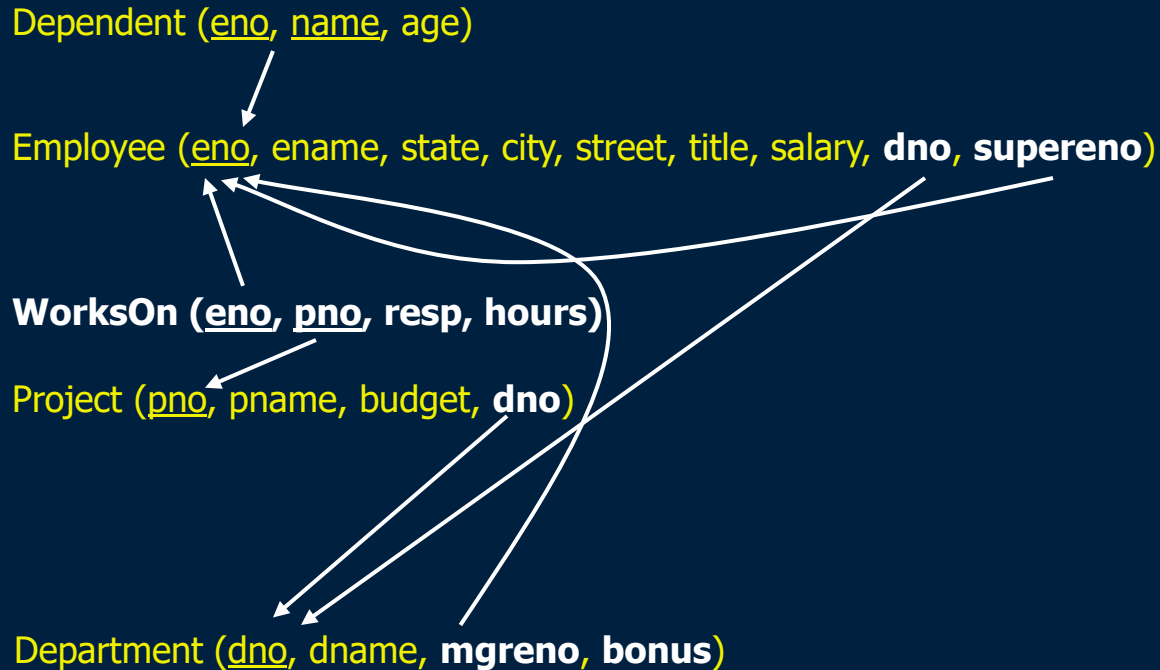


Step #5: Convert a binary M:N relationship into a new relation. Include the primary keys of both participating entity types as foreign keys; together they form the primary key. Add any relationship attributes.



ER to Relational Mapping

Current Relational Schema - Step #5

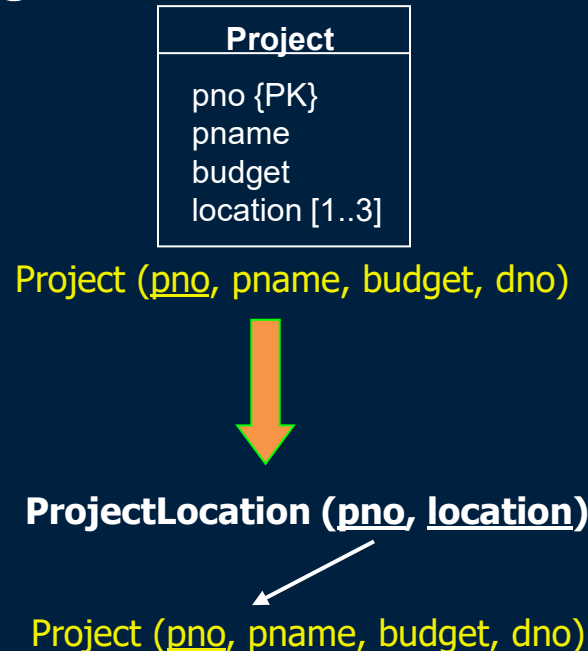


ER to Relational Mapping

Step #6: Convert Multivalued Attributes

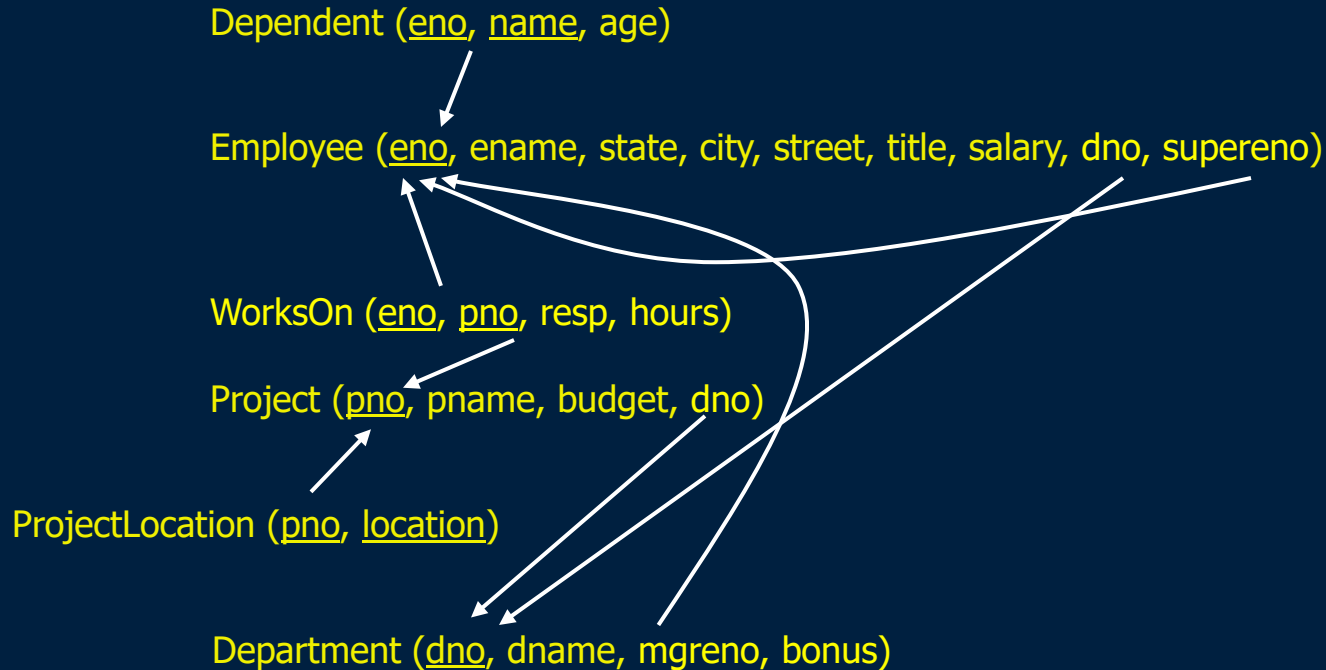


Step #6: Convert a multivalued attribute into a new relation with: the owner's primary key as a foreign key, the multivalued attribute, and a primary key consisting of both.



ER to Relational Mapping

Running Example

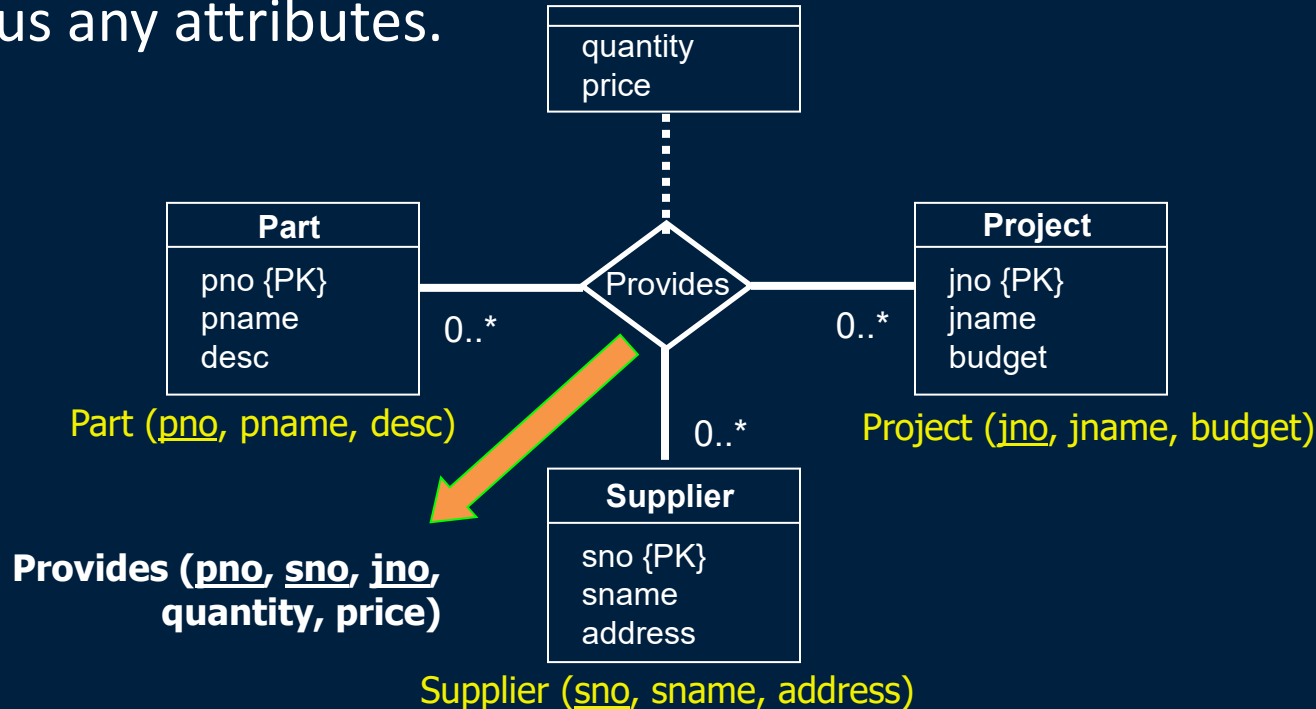


ER to Relational Mapping

Step #7: Convert n -ary Relationships



Step #7: Convert n -ary relationships by creating a new relation containing the primary keys of all participating entity types as foreign keys, plus any attributes.



EER to Relational Mapping

There are several ways to map superclass/subclass structures to relations. We will use the most general approach: one relation for the superclass and one for each subclass.

Approaches:

- 1) Create a separate relation for each superclass and subclass.
- 2) Create relations for subclasses only.
 - Only works if superclass has mandatory participation.
- 3) Create a single relation with one type attribute.
 - Attribute is used to indicate the type of object (subclass) in the row.
 - Works only if the subclasses are disjoint.
- 4) Create a single relation with multiple type attributes.
 - Have a boolean valued attribute for each subclass. True if in subclass.
 - Works if subclasses may be overlapping.

EER to Relational Mapping

Step #8: Convert Subclasses



Step #8: Create one relation for the superclass and one for each subclass.

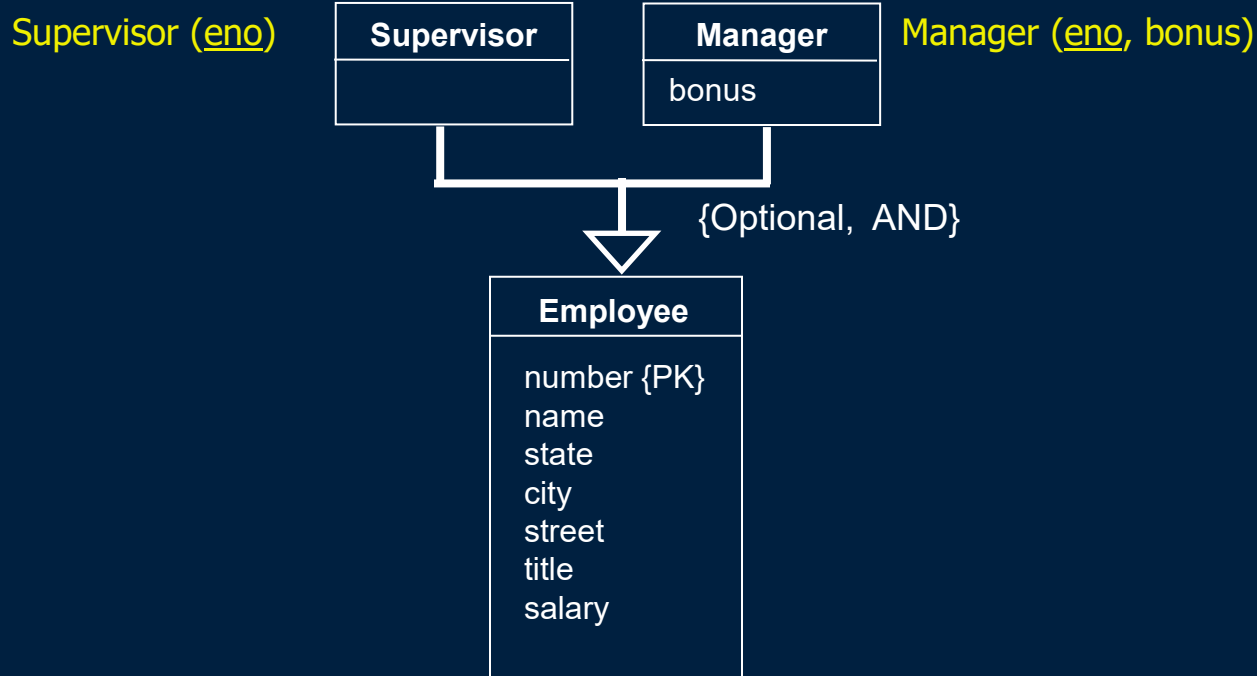
- Map the superclass to a relation.
- Each subclass gets its own relation containing its subclass-specific attributes.
- The superclass primary key becomes both the PK and an FK in each subclass relation.

EER to Relational Mapping

Step #8: Convert Subclasses



Step #8: Create a relation for each subclass and superclass. Link the subclasses to the superclass using foreign key references.



Employee (eno, ename, state, city, street, title, salary)

Summary of ER to Relational Mapping

ER Model

Strong Entity

Weak Entity

1:1 Relationship Type

1:N Relationship Type

M:N Relationship Type

n -ary Relationship Type

Multivalued attribute

Key attribute

Superclass/subclass

Relational Model

Relation

Relation; PK=owner PK(s)+partial key(s) if any
UNIQUE FK on one side

FK on N side referencing 1 side

New relation with two FKs

New relation with n FKs

New relation; owner PK + value form PK

Primary key attribute

Superclass relation + subclass relations;
subclass PK is also FK to superclass

ER to Relational Mapping Question

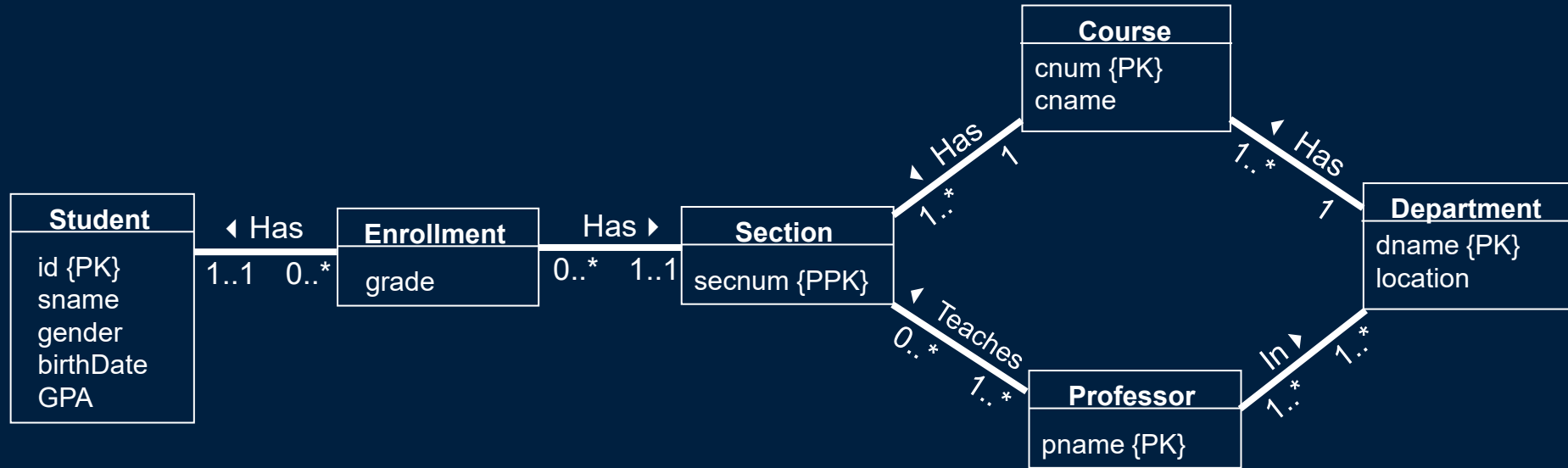
Question: How many of the following statements are **true**?

- 1) The M:N relationship mapping rule could be applied to 1:1 and 1:N relationships, as it is more general.
- 2) A M:N relationship is mapped to a relation in the relational model.
- 3) The designer has a choice on which side to put the foreign key when mapping a 1:N relationship.
- 4) When mapping a multivalued attribute, the new table containing the multivalued attribute will have a composite key.

A) 0 B) 1 C) 2 D) 3 E) 4

ER to Relational Mapping

University Question

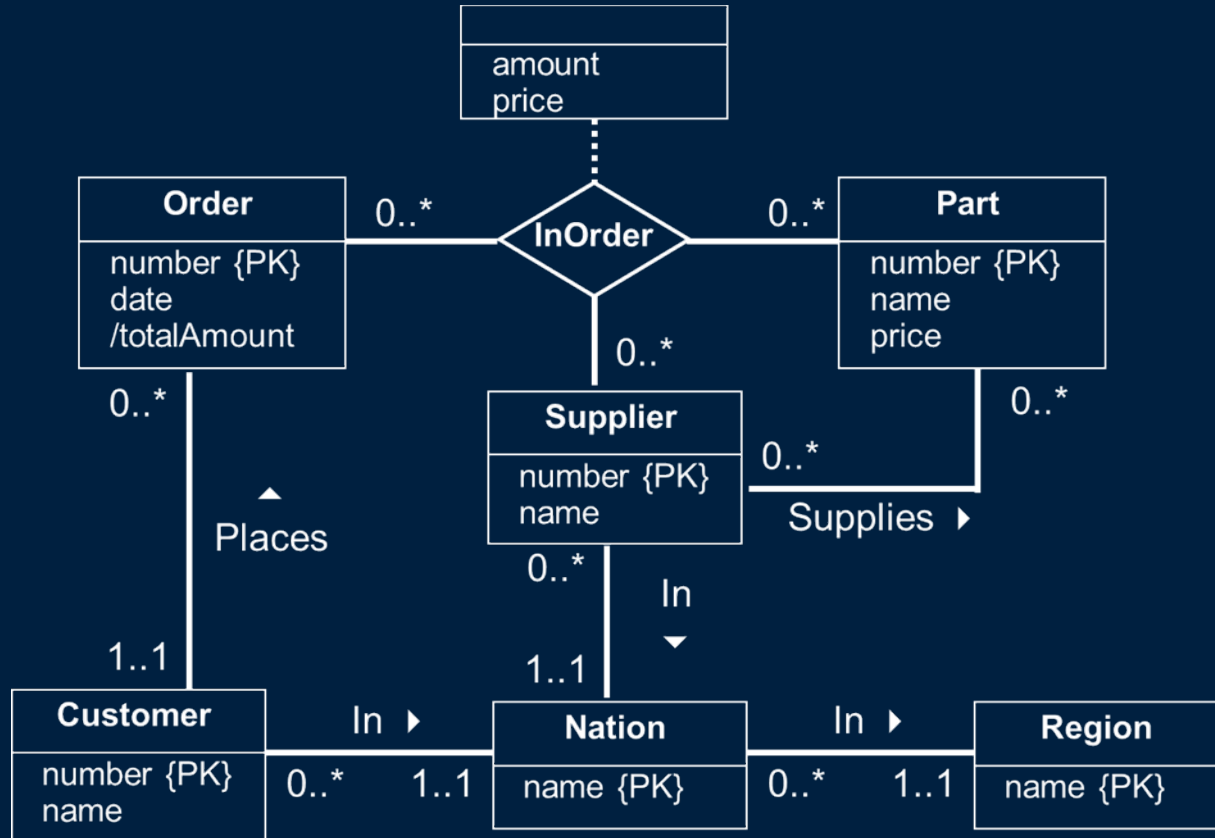


Question: Convert to the relational model.

Course notation: A weak entity has no complete PK of its own. A 1..1 relationship from the weak entity to another entity marks that entity as an identifying owner. With multiple identifying owners, their keys combine with any partial key to identify the weak entity. In this example, *Section* has identifying entity *Course*. *Enrollment* has identifying entities *Student* and *Section*.

ER to Relational Mapping

Simplified TPC-H ER Mapping Question



Conclusion

There is a straightforward algorithm for converting ER models to relational schemas.

The algorithm involves 7 steps for converting regular ER models, and 8 steps for converting EER models.

In general, entity types become relations. M:N and n-ary relationships become relations, while 1:1 and 1:N relationships are normally represented using foreign keys.

The mapping is systematic and can largely be automated.

Objectives

Given an ER/EER diagram, convert it into an equivalent relational schema.

Map superclass/subclass structures using relations linked by primary/foreign keys.



THE UNIVERSITY OF BRITISH COLUMBIA

